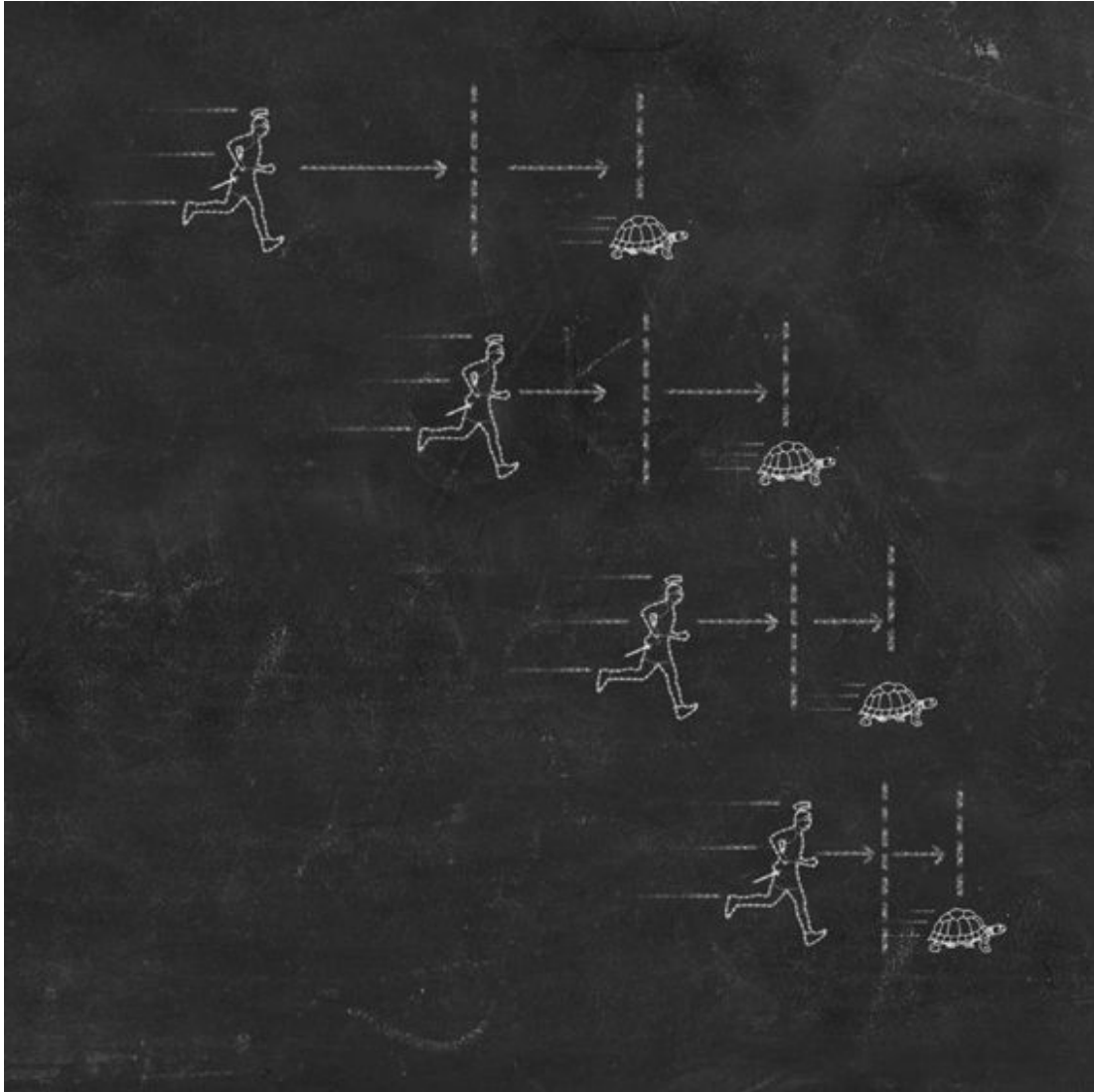


Zeno's paradoxes

Like his mentor Parmenides, Zeno of Elea (not to be confused with Zeno of Citium, see [Stoics: philosophy of the Roman Empire](#)) believed that 'all is one' and that change is impossible. To support his arguments, he used paradoxes – examples that seem to be logically sound but reach a conclusion that flies in the face of common sense. For example, Zeno pointed out that, at any instant in time, an arrow in flight is in one particular place and therefore not moving; and since time is made up of successive instants in which the arrow is motionless, the arrow cannot be moving.

A different problem of motion is tackled in the paradox of Achilles and the tortoise: in a race together, Achilles gives the tortoise a head start, and they both start running at their different speeds. When Achilles reaches the point where the tortoise started, the tortoise has advanced a shorter distance ahead; by the time Achilles has caught up to this point, the tortoise has moved further on, and so on – so Achilles can never catch up with the tortoise, let alone overtake him.



According to Zeno, the tortoise will always beat Achilles if it is given a head start.